

MET CS 555 – Foundation of Machine Learning

Module - 4

Lecture - 7: Regression Diagnostic

Lecture - 8: One-Way Analysis of Variance

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1. [Regression Diagnostics](#)

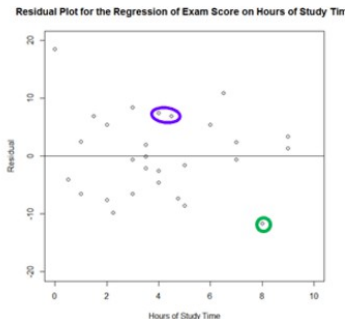
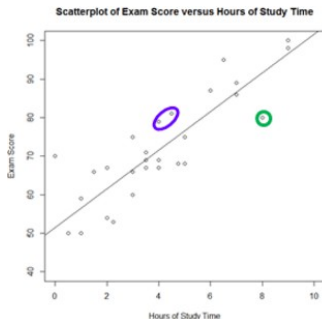
Regression Diagnostics

Regression Diagnostics

- ▷ **Determine whether all the necessary model assumptions are valid before performing inference.**
- ▷ If there are any violations, subsequent **inferential procedures may be invalid resulting in faulty conclusions.**
- ▷ In Regression Diagnostics we check if the assumptions are met in order to have confidence in the inferences.
- ▷ These techniques involve visualizing.
- ▷ We look for issues with **violations of the assumptions of the regression model.**

Residual Plots

- ▷ Residual plots help visualize how well a regression equation fits the sample data.
- ▷ Residual plots are scatterplots of the regression **residuals (y-axis)** against the **explanatory variable (x-axis)**.
- ▷ The residual plot **turns the regression line on the horizontal line** to see patterns and unusual observations.



Residual Plots

- ▷ Residual plots can also be generated using the **predicted values on the x-axis** as opposed to the explanatory variable (**especially more useful when MLR**).
- ▷ Residual plots can also be generated by **plotting standardized or studentized residuals**¹ (which involves dividing the residual by an estimate of the variability of the residuals).

¹Read more https://en.wikipedia.org/wiki/Studentized_residual
https://en.wikipedia.org/wiki/William_Sealy_Gosset

R commands - Residual Plots

```
# Calculates the residual values of a Linear Regression Model
> resid(m)

# Creating 4 plots side by side in 2 rows and 2 columns
> par(mfrow=c(2,2))
> plot(fitted(m), resid(m), axes=TRUE, frame.plot=TRUE, xlab='fitted
  values', ylab='residue')
> plot(age, resid(m), axes=TRUE, frame.plot=TRUE, xlab='age', ylab='
  residue')
> plot(height, resid(m), axes=TRUE, frame.plot=TRUE, xlab='height', ylab
  ='residue')

#
> hist(resid(m))
```

Outliers and Influence Points

Outliers:

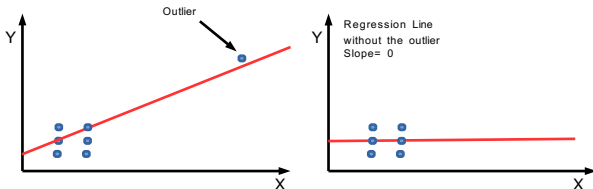
- ▷ Outliers are observations that lie outside the overall pattern of the other observations.
- ▷ Outliers can be identified via review of the scatterplot.
- ▷ Outliers in the y-direction tend to have large residuals.
- ▷ Outliers in the x-direction may or may not have large residuals but have **the potential to be influential**.

Influence Point:

- ▷ An influence point is an observation that **markedly changes the result of the regression if it were to be removed from the calculation**.

Outliers and Influence Points

- ▶ Least-squares regression is based on minimizing the squared vertical distances between the observations and the regression line (l2 norm), extreme points in the **x-direction tend to pull the regression line close to itself**.
- ▶ In these cases, the regression line equation may be **quite different with or without the points** and thus the points influences the regression equation.
- ▶ The influence of a particular point should be examined by removing it from the regression calculation and checking how the equations, inference, and conclusions change with its removal.



Outliers and Influence Points

- ▷ When there are outliers, we should **always check to ensure that there was not an issue with data entry/recording.**
- ▷ If an outlier in the x-direction, for example, is kept, it may be desirable **to collect additional data within the same range to better characterize the relationship** and so that the regression doesn't depend so heavily on the data from a single observation.

4 Principal Assumptions of the least-square regression

The following conditions must all be met before it is appropriate to make inference from a least-squares regression:

- ▷ The true relationship is **linear**.
- ▷ The observations are **independent**.
- ▷ The **variation** of the response variable around the regression line is **constant** (Constance Variance).
- ▷ The residuals are **normally distributed**.

If any of these are violated, then the inference, prediction and interpretation of the regression equation (or correlation) are inefficient (at best) or misleading/biased/incorrect (at worst).

Linearity and additivity of the relationship

Linearity and additivity of the relationship between dependent and independent variables:

- ▷ The expected value of dependent variable is a **straight-line function** of each independent variable, holding the others fixed.
- ▷ **Slope does not depend on the values** of the other variables.
- ▷ **The effects of different independent variables** on the expected value of the dependent variable **are additive**.

Regression Diagnostics

Check the assumptions:

- ▷ **Linearity**
- ▷ **Independence**
- ▷ **Constance variance**
- ▷ **Normally distributed residuals**

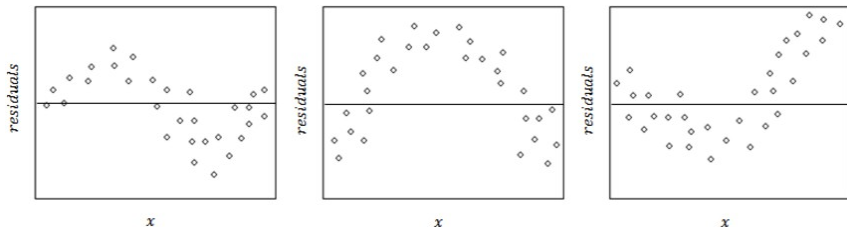
```
# Residual Plots (to assess linearity and constant variance
> plot([ variable for x-axis], resid(m))
# Check each explanatory variable, and the fitted values (fitted(m))
# the linearity or variance.

# Histograms (to check the distribution of the residuals)
> hist(resid(m))
```

Linearity

- ▷ Generate a scatterplot to visualize a roughly linear trend between factors.
- ▷ Be cautious of curved or other non-linear relationships.
- ▷ Residual plots can help assess the assumption as they can magnify non-linearity.
- ▷ **Violations of linearity or additivity are extremely serious:** your predictions might to be seriously in error when data are non-linearly or non-additively related.

Examples of Violations:



Independence

We make the assumption that the **observations are independent**.

Example:

Assume we are summarizing data on heights and weights in children, for example, that we only take **one observation per child** (as opposed to **multiple observations of the same child over time** or observations on various sets of **identical twins**).

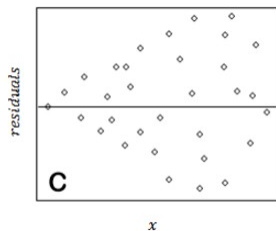
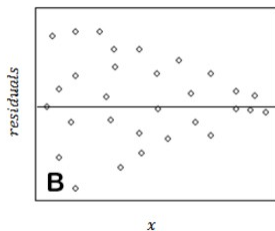
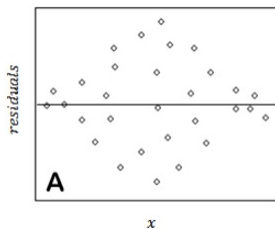
Correlated Data:

Observations on correlated data require more sophisticated analysis to account for the correlation between observations.

Constant Variance

- ▷ We assume that the variability of the response is constant across the regression line.
- ▷ This particular assumption can be checked via a residual plot.
- ▷ The residual plot should show approximately the same amount of scatter from left to right.

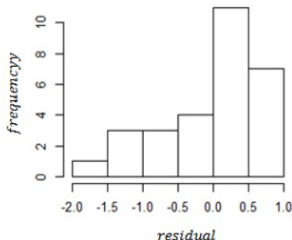
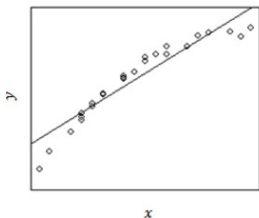
Examples of Violations:



Normality

- ▷ The residuals should follow a **normal distribution**.
- ▷ Severe deviations from this assumption could be **due to outliers or non-normality** of the explanatory or response variables.
- ▷ Residuals may not be normally distributed if the **linearity assumption has been violated**.
- ▷ Fortunately, **inference is not as sensitive** to departures from this assumption, especially when the number of observations is large.

To check this assumption use histograms of the residuals.



Standardized and Studentized Residuals

The residuals of a model (e_i) can tell us a lot about the model fit.

We generally do not study the residuals themselves, we study **a standardized form of the residuals**:

$$e_i^* = \frac{e_i - \bar{e}}{\sqrt{MSE}}$$

Where

- ▷ e_i is the residual for observation i and
- ▷ MSE is the mean squared error of residuals.

If $\sqrt{MSE}$ were an estimate of the standard deviation of the residual, we call e_i^* a studentized residual.

Residual Plots

```
> resid(m)
> par(mfrow=c(2,2))
> plot(fitted(m), resid(m), axes=TRUE, frame.plot=TRUE, xlab='fitted
  values', ylab='residue')
> plot(age, resid(m), axes=TRUE, frame.plot=TRUE, xlab='age', ylab='
  residue')
> plot(height, resid(m), axes=TRUE, frame.plot=TRUE, xlab='height', ylab
  ='residue')
> hist(resid(m))
```

fitted() is a generic function which extracts fitted values from objects returned by modeling functions.

Transformations

Assumptions of regression are violated?

Transformations can often help and be applied. If the variance of the response increases/decreases as the explanatory variable increases/decreases, then either

- ▷ **the natural log (ln)** or
- ▷ **the square root function** can be applied to the response variable to help “stabilize” the variance.

If there is a non-linear relationship between factors, then sometimes squaring the explanatory variable (or adding a squared term to a multiple linear regression model) can help.

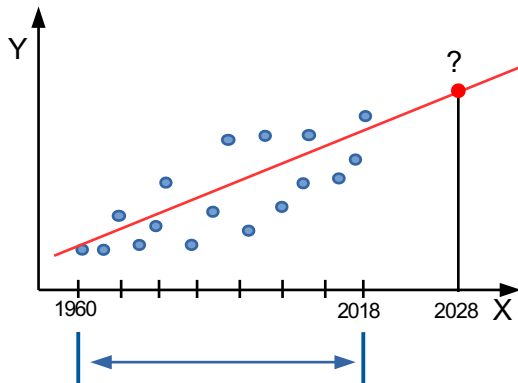
Finding the right transformation often involves some trial and error.²

If transformation **only marginally improves**, then **no transformation is preferred** (Due to the increased complexity)

²See section 11.10 and 11.1 in R Cookbook on transformations

Extrapolation

- ▷ Extrapolation is the **incorrect application of a regression equation outside of the range of data studied.**
- ▷ Interpolation produces estimates between known observations.
- ▷ Extrapolation is subject to **greater uncertainty and a higher risk of producing meaningless results.**



Lurking Variables

- ▷ Often there are variables that are **not measured but have strong influence on a dependent or an independent variable**. Such variables are called **lurking variables** (aka confounding).
- ▷ You should always consider this when interpreting regression results.

Example: Consider the correlation between study time and grade score.

Many other hidden variables might have an impact on final grade score.

Variables like **stress level of a person, amount of sleep per day, motivation score, etc ...**

Causation and Association

- ▷ Linear regression is useful to understand and quantify **associations** - which is **not the same thing as determining causation**.
- ▷ The regression shows changes in the explanatory variables are associated with changes in the response variable, it **is not the same as causing changes in the response variable**.

Example:

Amount of milk consumed in a country might be correlated to number of forest fires.

- But it does not mean that it caused that.
- Sometimes you might find a third factor that connects things, like weather.

It is easy to find correlation but it is hard to proof causation.

Causation and Association

- ▷ To show causation, an experiment where you **change the independent variable in specific ways and observe the resulting effect** on the response is the most convincing way to evaluate it.
- ▷ For an experiment, we **actually try to provoke a response by doing some actions**.
- ▷ Experiments are often expensive and not always ethical to conduct (Example medical research)
- ▷ **If an experiment cannot be performed**, then in order to begin to suggest that a causal relationship may apply, we'd want to see that the association was strong, the association is consistent across many different studies, the cause precedes the effect temporally, and the cause is plausible (scientifically and practically).

Multicollinearity

- ▷ When two or more independent **variables are highly correlated**, entering both of them into a multiple linear regression model **may be problematic as the effect of each may cancel the other out.**
- ▷ This phenomenon is called **collinearity or multicollinearity**.
Multicollinearity increases the standard errors of the coefficients.
- ▷ Increased SE means some **regression coefficients are close to 0**.
Multicollinearity makes some variables statistically insignificant when they should be significant.

Multicollinearity

What to do?

- ▷ Look at each independent variable's **association with the dependent variable separately before looking at them together** as well as examining the **relationship between independent variables**.
- ▷ If two independent variables are highly correlated (*correlation* > 0.8) then it may be advisable to only **select one for inclusion in the regression**.

Warning Signs of Multicollinearity

Severe multicollinearity is a major problem because it increases the variance of the regression coefficients, making them unstable.

Here are some things to watch for:

1. **A regression coefficient is not significant** even though, theoretically, that variable should be highly correlated with Y.
2. When you **add or delete an X variable**, the **regression coefficients change dramatically**.
3. You see a **negative regression coefficient** when your **response should increase** along with X.
4. You see a **positive regression coefficient** when the **response should decrease** as X increases.
5. Your X variables have **high pairwise correlations**.

Check the linear model you built

▷ **Is the model statistically significant?**

Check the F statistic (at the bottom of the summary)

▷ **Are the coefficients significant?**

Check the coefficients t statistics and p-values in the summary, or check their confidence intervals

▷ **Is the model useful?**

Check the R^2 near the bottom of the summary

▷ **Does the model fit the data well?** Plot the residuals and check the regression diagnostics

▷ **Does the data satisfy the assumptions behind linear regression?**

Check if the diagnostics confirm that a linear model is reasonable for your data

Real-World Applications

▷ **Economics and Finance:**

Predicting Consumer Behavior: Estimating how factors like income and price influence consumer spending.

Stock Market Analysis: Modeling the relationship between market indicators and stock prices.

▷ **Healthcare and Medicine:**

Drug Efficacy Studies: Assessing how dosage levels affect patient recovery rates.

Epidemiology: Investigating the relationship between environmental factors and disease incidence.

▷ **Environmental Science:**

Climate Modeling: Predicting temperature changes based on greenhouse gas emissions.

Pollution Analysis: Estimating pollution levels from industrial activity data.

▷ **Engineering:**

Quality Control: Predicting product failure rates based on manufacturing variables.

Structural Analysis: Modeling stress-strain relationships in materials.

▷ **Marketing and Business:**

Sales Forecasting: Estimating future sales based on advertising spend and market trends.

Customer Relationship Management: Understanding how customer satisfaction impacts loyalty.

Common Mistakes and Pitfalls

▷ **Assuming Linearity Unconditionally:**

Mistake: Applying linear regression to data with a nonlinear relationship.

Solution: Perform exploratory data analysis (EDA) to visualize relationships before modeling.

▷ **Ignoring Regression Assumptions:**

Mistake: Overlooking assumptions like homoscedasticity (constant variance of errors), independence, and normality of residuals.

Solution: Use diagnostic plots (e.g., residual plots, Q-Q plots) to check assumptions.

▷ **Multicollinearity in Multiple Regression:**

Mistake: Including highly correlated independent variables, which can distort coefficient estimates.

Solution: Calculate Variance Inflation Factor (VIF) to detect multicollinearity and consider removing or combining correlated variables.

▷ **Overfitting the Model:**

Mistake: Including too many variables, leading the model to fit the noise rather than the signal.

Solution: Use techniques like cross-validation and adjust for complexity using criteria like AIC or BIC.

Common Mistakes and Pitfalls (Continued)

▷ **Underfitting the Model:**

Mistake: Using too simple a model that doesn't capture the underlying data patterns.

Solution: Incorporate additional relevant variables and consider interaction terms.

▷ **Extrapolation Beyond Data Range:**

Mistake: Making predictions outside the range of the observed data where the model may not be valid.

Solution: Limit predictions to the range of the data or use models better suited for extrapolation.

▷ **Misinterpreting Correlation as Causation:**

Mistake: Assuming that a relationship implies causality without experimental or longitudinal evidence.

Solution: Be cautious in interpretation and, if possible, use randomized controlled trials to establish causation.

▷ **Ignoring Outliers and Influential Points:**

Mistake: Not identifying or addressing outliers, which can skew results.

Solution: Analyze residuals to detect outliers and consider robust regression methods or data transformation.

Effective Use of Linear Regression

▷ **Appropriate Contexts:**

When there is a plausible linear relationship between variables.

For predictive modeling within the scope of available data.

When the primary goal is to understand associations rather than causation.

▷ **Best Practices:**

Data Preprocessing: Clean data thoroughly, handle missing values, and encode categorical variables properly.

Assumption Verification: Always check the underlying assumptions of linear regression.

Model Validation: Use training and testing datasets or cross-validation to assess model performance.

Interpretation: Carefully interpret coefficients, considering the effect size and statistical significance.

When to Avoid Linear Regression

▷ **Nonlinear Relationships:**

When data exhibit curvature that linear models cannot capture.

Alternative: Use polynomial regression or other nonlinear models.

▷ **Violation of Assumptions:**

When residuals show patterns indicating heteroscedasticity or autocorrelation.

Alternative: Consider transforming variables or using generalized linear models.

▷ **High Dimensionality with Limited Data:**

When the number of predictors is close to or exceeds the number of observations.

Alternative: Use dimensionality reduction techniques or regularization methods like Lasso or Ridge regression.

▷ **Causal Inference without Proper Design:**

When trying to infer causality from observational data without controlling for confounding variables.

Alternative: Use experimental designs or advanced methods like instrumental variables.

When to Avoid Linear Regression (Continued)

▷ What Is Causal Inference Without Proper Design?

- Causal Inference:** The process of determining whether one variable (the cause) directly affects another variable (the effect).
- Proper Design:** A study design that adequately controls for confounding variables and biases, often through randomization or experimental manipulation.
- Mistake:** Using regression analysis on observational data to claim that changes in the independent variable cause changes in the dependent variable, without accounting for other factors that might influence the relationship.
- Why It's a Problem:** Observational data often contain confounding variables—factors that affect both the independent and dependent variables—leading to spurious or misleading associations.

What is Confounding Variables?

- Confounding Variables:** Variables that the researcher failed to control or eliminate, damaging the internal validity of an experiment.
- Effect:** They provide alternative explanations for an observed relationship, making it difficult to establish a direct cause-and-effect link.

Example

Suppose you find a positive correlation between coffee consumption and heart disease. You might be tempted to conclude that coffee causes heart disease. However, if you fail to account for smoking habits (a confounding variable), which are associated with both higher coffee consumption and increased heart disease risk, your conclusion would be flawed.

3D plots

```
> install.packages("rgl")
```

rgl Provides medium to high level functions for 3D interactive graphics, including functions modelled on base graphics (plot3d(), etc.) as well as functions for constructing representations of geometric objects (cube3d(), etc.). Output may be on screen using OpenGL, or to various standard 3D file formats including WebGL, PLY, OBJ, STL as well as 2D image formats, including PNG, Postscript, SVG, PGF.

```
> library(rgl)
```

```
> plot3d(age, height, salary, type = "s", size = .75, xlab="Age", ylab="Height", zlab="Annual Salary")
```

MET CS 555 – Foundation of Machine Learning

Module - 4

Lecture - 8: One-Way Analysis of Variance

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One-Way Analysis of Variance

One-Way Analysis of Variance

- ▷ Analysis of variance is a general term which involves breaking down the **overall variability in a particular continuous outcome into pieces.**
- ▷ It involves comparing the variability after accounting for a characteristic versus the remaining variability not explained by the characteristic and just inherent to the outcome.
- ▷ In **one-way analysis of variance (ANOVA)**, we study groups that are defined based on the value of **one factor.**
- ▷ **The goal of a one-way ANOVA is to compare means across groups.**
- ▷ In ANOVA framework, we make comparisons **across several groups** while considering all of the data together.

One-Way Analysis of Variance

To compare data across multiple groups, we will test the null hypothesis that the underlying population means are all equal versus the alternative that at least two of the underlying population means differ.

If we have **k groups** and we denote as the true population **mean for group i**, then the hypotheses for the **one-way ANOVA** can be written as follows:

▷ **$H_0 : \mu_1 = \mu_2 = \dots = \mu_k$** (All underlying population means are equal)

▷ **$H_1 : \mu_1 \neq \mu_i$**

(at least two of the underlying population means are different or not all of the underlying population means are the same/equal)

Example - One-Way Analysis of Variance

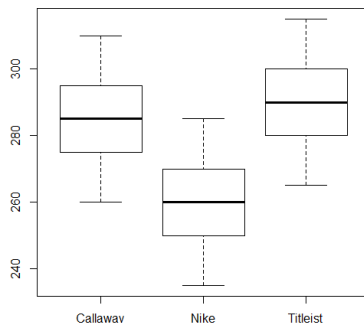
Commercials aired on TV entice potential buyers to consider purchasing specific brand of golf balls by claiming increased driving distances. In order to see which brand of golf balls is best (as measured by distance travelled), an experiment is set up where a mechanical driver hits 5 balls of each of 3 brands. The distance in yards achieved after each strike is measured.

Example - One-Way Analysis of Variance

Commercials aired on TV entice potential buyers to consider purchasing specific brand of golf balls by claiming increased driving distances. In order to see which brand of golf balls is best (as measured by distance travelled), an experiment is set up where a mechanical driver hits 5 balls of each of 3 brands. The distance in yards achieved after each strike is measured.

Table A. Distance by golf ball brand

Observation	Brand		
	Callaway	Nike	Titleist
1	275	235	265
2	310	285	300
3	285	270	280
4	260	250	315
5	295	260	290
Mean	285	260	290
Standard Deviation	19.0	19.0	19.0



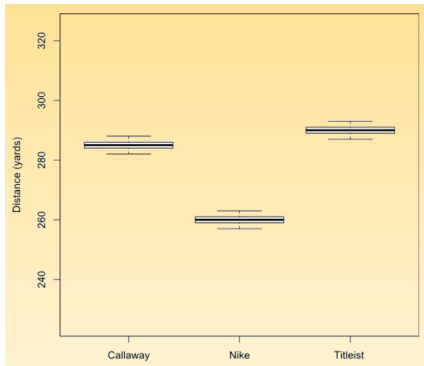
Example - One-Way Analysis of Variance

In both cases, the sample means for each of the brands is the same. The difference between the two versions of the examples are the amount of variability in the outcome.

When the variability is small relative to the differences between means, we become increasingly likely to declare that groups differ from each other.

Table B. Distance by golf ball brand

Observation	Brand		
	Callaway	Nike	Titleist
1	288	257	291
2	286	263	293
3	284	261	287
4	282	260	289
5	285	259	290
Mean	285	260	290
Standard Deviation	2.2	2.2	2.2



One-Way Analysis of Variance

If the **variability between groups**

- ▷ **is small** relative to the variability in the measurements **within groups**, we are less inclined to conclude that there **is a difference between them**.
- ▷ **is large** in comparison to the variability **within each group**, it is easier to see and conclude that **there is a difference**.

In ANOVA, we use the F-statistic for this test where the F-statistic is calculated as

$$\begin{aligned} F &= \frac{s_b^2}{s_w^2} \\ &= \frac{\text{between group variance}}{\text{within group variance}} \\ &= \frac{\text{mean square between}}{\text{mean square within}} \end{aligned}$$

One-Way Analysis of Variance

The numerator of F is an estimate of the between group variation. It shows how far the group means are from the overall mean (across all groups).

$$\begin{aligned}s_b^2 &= \text{mean square between} \\ &= \frac{\text{SSB}}{k - 1} \\ &= \frac{\text{sum of squares between}}{\text{number of groups} - 1} \\ &= \frac{\sum_{j=1}^k n_j (\bar{x}_{.j} - \bar{x}_{..})^2}{k - 1}\end{aligned}$$

where

n_j is the number of observations in group j ,

▷ $\bar{x}_{.j}$ is the sample mean for group j ,

$\bar{x}_{..}$ is the overall mean (using all observations across all groups), and

▷ k is the number of groups.

One-Way Analysis of Variance

The denominator of F is an estimate of the within group variation. How far each individual data point is from the corresponding groups mean and take a weighted average.

$$\begin{aligned}s_w^2 &= \text{mean square within} \\ &= \frac{SSW}{n - k} \\ &= \frac{\text{sum of squares within}}{\text{number of observations} - \text{number of groups}} \\ &= \frac{\sum \sum (x_{ij} - \bar{x}_{.j})^2}{n - k} \\ &= \frac{\sum_{j=1}^k (n_{.j} - 1)s_j^2}{n - k}\end{aligned}$$

where

- ▷ n is the number of observations across all groups,
- x_{ij} is the i th observation in the group j ,
- ▷ $\bar{x}_{.j}$ is the sample mean for group j ,
- $n_{.j}$ is the number of observations in group j , and
- ▷ k is the number of groups.

One-Way Analysis of Variance an example

Observation	Brand		
	Callaway (1)	Nike (2)	Titleist (3)
1	280	260	280
2	275	255	290
3	290	270	295
4	295	265	300
5	285	250	285
Mean	285	260	290
Standard Deviation	7.9	7.9	7.9
Variance	62.5	62.5	62.5

One-Way ANOVA An Example - Mean Square Between

$$\begin{aligned}\text{mean square between} &= s_b^2 \\&= \frac{\text{sum of squares between}}{\text{number of groups} - 1} \\&= \frac{SSB}{k - 1} \\&= \frac{\sum_{j=1}^k n_{.j}(\bar{x}_{.j} - \bar{x}_{..})^2}{k - 1} \\&= \frac{5 \cdot (285 - 278.33)^2 + 5 \cdot (260 - 278.33)^2 + 5 \cdot (290 - 278.33)^2}{3 - 1} \\&= \frac{222.4445 + 1679.9445 + 680.9445}{2} \\&= \frac{2583.3335}{2} \\&= 1291.6668\end{aligned}$$

One-Way ANOVA An Example - mean square within

mean square within

$$= \frac{\text{sum of squares within}}{\text{number of observations} - \text{number of groups}}$$

$$= \frac{SSW}{n - k}$$

$$= \frac{\sum \sum (x_{ij} - \bar{x}_{.j})^2}{n - k}$$

$$= \frac{(280 - 285)^2 + (275 - 285)^2 + . + (285 - 285)^2 + . + (280 - 290)^2 + (290 - 290)^2 + . + (285 - 290)^2}{15 - 3}$$

$$= \frac{750}{12}$$

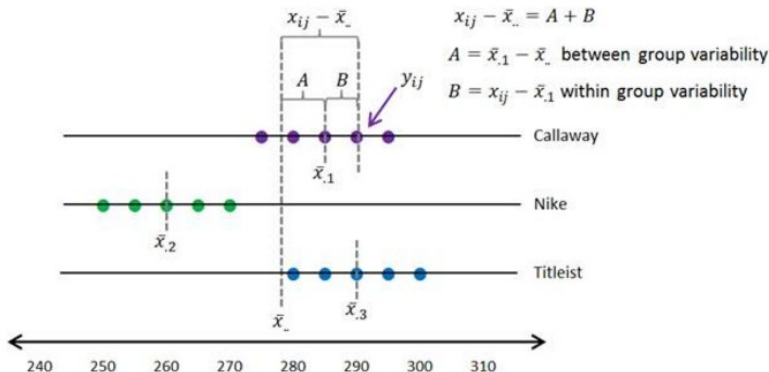
$$= 62.5$$

One-Way ANOVA

The deviation between an individual point and the overall mean is comprised of two parts:

1. deviation between the individual point and the respective group mean,
2. deviation between the group mean and the overall mean

$$x_{ij} - \bar{x}_{..} = x_{ij} - \bar{x}_{.j} + \bar{x}_{.j} - \bar{x}_{..} = (x_{ij} - \bar{x}_{.j}) + (\bar{x}_{.j} - \bar{x}_{..})$$



One-Way Analysis of Variance

- ▷ The deviation between the individual point and the respective group mean is representative of the **within group variability**. The deviation between the group mean and the overall mean is representative of **the between group variability**.
- ▷ If the between group variability **is large** and the within group variability **is small**, then the **null hypothesis is often rejected** (in favor of the alternative hypothesis that the underlying means across group are not all equal).

One-Way Analysis of Variance

- ▷ **Large values of the F-statistic** indicate that the variation between groups is larger than the variation within each individual group.
- ▷ To know how large is large enough to **reject the null hypothesis**, we use the **F-distribution with $k-1$ and $n-k$ degrees of freedom**.

Inference - F-Test - Global Test

The F-test derived from the ANOVA table is sometimes referred to as the **global test**.

The global F-test only has the ability to conclude that **there are group differences (if the null hypothesis is rejected)**, but **does not allow one to know which groups are different** without taking further steps to investigate where the differences lie.

	SS (Sum of Squares)	df (degrees of freedom)	MS (Mean Square)	F-statistic	p-value
Between	SSB	SSB df = $k - 1$	$MSB = SSB / \text{SSB df} = s_b^2$	$F = s_b^2 / s_w^2$	$P(F_{\text{SSB df, SSW df}, \alpha} > F)$
Within	SSW	Res df = $n - k$	$MSW = SSW / \text{SSW df} = s_w^2$		
Total	Total SS = SSB + SSW				

Inference - Anova table

	SS (Sum of Squares)	df (degrees of freedom)	MS (Mean Square)	F-statistic	p-value
Between	SSB	SSB df = $k - 1$	$MSB = SSB / \text{SSB df} = s_b^2$	$F = s_b^2 / s_w^2$	$P(F_{\text{SSB df, SSW df}, \alpha} > F)$
Within	SSW	Res df = $n - k$	$MSW = SSW / \text{SSW df} = s_w^2$		
Total	Total SS = SSB + SSW				

$$SSB = \sum_{j=1}^k n_{.j} (\bar{x}_{.j} - \bar{x}_{..})^2$$

$$SSW = \sum \sum (x_{ij} - \bar{x}_{.j})^2 = \sum_{j=1}^k (n_{.j} - 1) s_j^2$$

$$\text{Total SS} = \sum \sum (x_{ij} - \bar{x}_{..})^2$$

$$F = MSB / MSW = s_b^2 / s_w^2$$

F-test for ANOVA: An Example

A random sample ($n=19$) of current light smokers, current heavy smokers, former smokers, and those who have never smoked was taken to determine if mean systolic blood pressure (SBP) differs across smoking status categories.

1. Set up the hypotheses and select the alpha level

$$H_0 : \mu_{heavy} = \mu_{light} = \mu_{former} = \mu_{never}$$

(All underlying population means are equal)

$$H_1 : \mu_i \neq \mu_j \text{ for some } i \text{ and } j.$$

(Not all of the underlying population means are equal)

$$\alpha = 0.05$$

2. Select the appropriate test statistic

$$F = \frac{MSB}{MSW} \text{ with } k - 1 = 3 \text{ and } n - k = 19 - 4 = 15 \text{ degrees of freedom}$$

F-test for ANOVA: An Example (continued)

3. State the decision rule F-distribution with 3, 15 degrees of freedom and associated with $\alpha = 0.05$.

> **qf** (.95 , df1 =3, df2 =15)
> F3 ,15 ,0.05=3.287

Decision Rule: Reject H_0 if $F \geq 3.287$

Otherwise, do not reject H_0

4. Compute the test statistic

$$F = \frac{MSB}{MSW} = \frac{928.7}{43.2} = 21.49$$

5. Conclusion

Reject H_0 since $21.49 \geq 3.287$.

We have significant evidence at the $\alpha = 0.05$ that there is a difference in SBP among current light smokers, current heavy smokers, former smokers, and those who have never smoked

F-test for ANOVA: An Example (continued) - Using R

```
> data <- read.csv("smoking_SBP.csv")

# Check if the variable group is factor variable in R.
# If not make it a factor variable

> is.factor(data$group)

# aov(data$response~data$group)
> m <- aov(data$SBP~data$group, data=data)

# pass the anova model object to the summary function.
> summary(m)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
group	3	2786.2	928.7	21.49	1.1e-05 ***
Residuals	15	648.3	43.2		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Pairwise Comparisons

Evaluating group differences - t-test

If the global **F-test is significant**, then it is of interest to further determine which of the population group means are different.

We perform testing on each **pairwise comparison of interest**.

We use a t statistic:

$$t = \frac{\bar{x}_i - \bar{x}_j}{\sqrt{s_p^2 \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}}$$

- ▷ $\bar{x}_i$ are $\bar{x}_j$ the sample mean in groups i and j (respectively),
- ▷ s_p^2 is the estimate of the variance from the ANOVA model (and is equal to the mean square within, or s_w^2),
- ▷ n_i and n_j are the number of observations in groups i and j (respectively), which follows a t-distribution with $n - k$ degrees of freedom under H_0 .

Evaluating group differences - t-test

The decision rule for a two-sided level α test is:

- ▷ **Reject $H_0 : \mu_i = \mu_j$ if $t \geq t_{n-k, \frac{\alpha}{2}}$ or if $t \leq -t_{n-k, \frac{\alpha}{2}}$**
- ▷ Otherwise, do not reject $H_0 : \mu_i = \mu_j$

where $t_{n-k, \frac{\alpha}{2}}$ is the value from the t-distribution table with $n - k$ degrees of freedom and associated with a right hand tail probability of $\alpha/2$.

Evaluating group differences - t-test - Confidence Interval

We can also calculate the two-sided $100\% \times (1 - \alpha)$ confidence interval for the difference between means ($\mu_i - \mu_j$) using the following formula:

$$(\bar{x}_i - \bar{x}_j) \pm t_{n-k, \frac{\alpha}{2}} \cdot \sqrt{s_p^2 \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}$$

We can say with $100\% \times (1 - \alpha)$ confidence that difference between the underlying means of groups i and j is between the above values.

Example: t-test

The golf ball example.

1. Set up the hypotheses and select the alpha level

$$H_0 : \mu_{\text{Titleist}} = \mu_{\text{Callaway}}$$

$$H_1 : \mu_{\text{Titleist}} \neq \mu_{\text{Callaway}}$$

$$\alpha = 0.05$$

2. Select the appropriate test statistic

$$t = \frac{\bar{x}_{\text{Titleist}} - \bar{x}_{\text{Callaway}}}{\sqrt{s_p^2 \left(\frac{1}{n_{\text{Titleist}}} + \frac{1}{n_{\text{Callaway}}} \right)}}$$

3. State the decision rule

Determine the appropriate value from the t-distribution table with $n-k=15-3=12$ degrees of freedom and associated with a right hand tail probability of $\alpha/2 = 0.05/2 = 0.025$

$$t_{n-k, \frac{\alpha}{2}} = t_{12, 0.025} = 2.179$$

```
> qt(.975, df=12)
> t12,0.025=2.179
```

Example: t-test (continued)

4. Compute the test statistic

$$\begin{aligned} t &= \frac{\bar{x}_{\text{Titleist}} - \bar{x}_{\text{Callaway}}}{\sqrt{s_p^2 \left(\frac{1}{n_{\text{Titleist}}} + \frac{1}{n_{\text{Callaway}}} \right)}} \\ &= \frac{290 - 285}{\sqrt{62.5 \left(\frac{1}{5} + \frac{1}{5} \right)}} = \frac{5}{\sqrt{25}} = \frac{5}{5} = 1 \end{aligned}$$

5. Conclusion

Do not reject H_0 since $1 < 2.179$.

We do not have significant evidence at the $\alpha = 0.05$ level that

$\mu_{\text{Titleist}} \neq \mu_{\text{Callaway}}$

That is, we do not have evidence that the mean distances are different between Titleist golf balls and Callaway golf balls ($p = 0.34$ as calculated using a software program).

Example: t-test (continued)

$$\begin{aligned}(\bar{x}_{\text{Titleist}} - \bar{x}_{\text{Callaway}}) \pm t_{n-k, \frac{\alpha}{2}} \cdot \sqrt{s_p^2 \left(\frac{1}{n_{\text{Titleist}}} + \frac{1}{n_{\text{Callaway}}} \right)} \\= (290 - 285) \pm 2.18 \cdot \sqrt{62.5 \left(\frac{1}{5} + \frac{1}{5} \right)} \\= 5 \pm 2.18 \cdot 5 \\= (-5.9, 15.9)\end{aligned}$$

We are 95% confident that the true difference between the two is between 5.9 yards (favoring the Callaway balls) and 15.9 yards (favoring the Titleist balls).

Example: t-test R command

```
> golf <- read.csv("golfball_C.csv")
> golf
> attach(golf)
> aggregate(dist, by=list(brand), summary)
> aggregate(dist, by=list(brand), var)

> pairwise.t.test(dist, brand, p.adj='none')
```

Pairwise comparisons using **t** tests with pooled SD

data: dist and brand

	Callaway	Nike
Nike	0.00031	-
Titleist	0.33705	6.2e-05

P value adjustment method: none

Type I and type II errors

Type I and type II errors

Type I Error: rejecting the null hypothesis when it is true (false positive)

- ▷ The probability of making a **Type I Error** is **controlled by the significance level of the test**. It is generally the error that we worry the most about.
- ▷ The potential implications of this type of error explains **why we specifying small values of alpha** (the significance level).
- ▷ The potential implications of this type of error necessitates the **need to report p-values in your summary of results**.

		Reality (unknown in practice)	
		H_0 is true	H_0 is false
Decision based on sample	Reject H_0	Type I Error	Correct Decision
	Fail to reject H_0	Correct Decision	Type II Error

Type I and type II errors

- ▷ **Type II Error:** failing to reject the null hypothesis when it is not true (false negative)
- ▷ The probability of making a **Type II Error** is controlled by the sample size. The probability decreases with increasing sample size.
- ▷ The **Type II Error** is known as the **power of a test** (the probability of rejecting the null hypothesis when the null hypothesis is not true).

Type I and type II errors

- ▷ When we **fail to reject the null hypothesis**, it is often **difficult to know whether or not we didn't have enough power** or if the null hypothesis was indeed true.
- ▷ **This is why we don't say we accept the null hypothesis** and we instead state our conclusion as failing to reject the null hypothesis or we do not have sufficient evidence to reject the null hypothesis.
- ▷ Since the probability of **type II of error decreases as the sample size increases**, type II errors rates of 10% or less are often selected for such settings.
In most other cases, a type II error rate of 20% is generally considered reasonable for planning purposes.

Issues with multiple comparisons

If there are k groups, the number of possible pairwise comparisons is:

$$c = k \cdot (k - 1) / 2$$

- ▷ If each of the possible $k \cdot (k - 1) / 2$ pairwise comparisons are performed at a significance level of α , then the expected number of false positives increases (up to as many as $\alpha \cdot c$).
- ▷ To maintain strong **control of the error rate** across all pairwise comparisons (often referred to as the experiment wise or family wise error rate) at a selected level, the significance level of each individual test needs to be adjusted.
- ▷ Procedures for controlling the family wise **type I error** rate at a pre-specified level are called multiple comparison procedures.
- ▷ These procedures work by essentially making it **“harder” to find differences between groups**. More evidence against the null hypothesis is needed to reject the null hypothesis when these procedures are implemented.

Issues with multiple comparisons - Bonferroni adjustment

Bonferroni adjustment: one of the simplest and most commonly used multiple comparisons procedures

To control the family wise error rate at the level, individual tests are performed at the $\alpha^* = \alpha / c$ level of significance

where

- ▷ c is the number of individual comparisons to be performed
- ▷ α is the significance level of each individual test.

The **family wise error rate** is calculated using the formula

$$1 - (1 - \alpha) \cdot c$$

Tukey procedure

The Studentized Range test or **Tukeys Test of Honest Significance Test** is another common multiple comparisons procedure which is commonly used to control the family wise **type I error rate for pairwise comparisons**.

It tends to **have more statistical power (is less conservative) than the Bonferroni method**, especially when there are a **large number of pairwise comparisons**.

The procedure is generally implemented using statistical software like R.

Issues with multiple comparisons: the golf example

In the golf ball example, the global F-test showed that there was a difference in mean distance between brands.

Then we can perform three pairwise comparisons (*Titleist versus Callaway, Titleist versus Nike, and Callaway versus Nike*).

However, we did not account for the fact that we were doing all three comparisons and did each at the $\alpha = 0.05$ level when really we had wanted **to control the family wise type I error rate at $\alpha = 0.05$ overall**.

The Bonferroni methodology suggests that individual tests should be performed at the $\alpha^* = \alpha/c$ level of significance,

$$\alpha^* = \alpha/c = 0.05/3 \simeq 0.0167$$

The critical value that we should have used in each comparison should have been

$$t_{n-k, \alpha^*/2} = t_{12, 0.00833} = 2.78$$

instead of $t_{12, 0.025} = 2.18$.

Example SBP by smoking status

```
# Check if grouping variable (smoking status) is a factor
> is.factor(data$group)

# Numerical and graphical summaries (Module 1 and 2)
# Calculate mean, SD of SBP by groups
# Box plots and histograms
> aggregate(data$SBP, by=list(data$group), summary)
> aggregate(data$SBP, by=list(data$group), sd)
> boxplot(data$SBP~data$group, data=data, main="SBP by smoking status",
  xlab="group", ylab="SBP", ylim=c(100, 160))

# Perform one-way ANOVA and if necessary, calculate the associated
  pairwise comparisons

> m<- aov(data$SBP~data$group, data=data)
> summary(m)

# bonferroni adjustment
> pairwise.t.test(data$SBP, data$group, p.adj="bonferroni")

# Tukeys Test of Honest Significance Test
> TukeyHSD(m)
```

The golf example R command

Pairwise comparisons using t tests

```
> pairwise.t.test(dist, brand, p.adj='none')
Pairwise comparisons using t tests with pooled SD
data:  dist and brand
      Callaway Nike
Nike    0.00031  -
Titleist 0.33705  6.2e-05
P value adjustment method: none
```

P value adjustment method: bonferroni

```
> pairwise.t.test(dist, brand, p.adj='bonferroni')

Pairwise comparisons using t tests with pooled SD
data:  dist and brand
      Callaway Nike
Nike    0.00093  -
Titleist 1.00000  0.00019
P value adjustment method: bonferroni
```

The golf example R command

Create Anova Model and getting Summary results

```
# golfbal example - distances and different brands

> m <- aov(dist ~ brand, data=golf)
> summary(m)

# Tukey s Test of Honest Significance Test
> TukeyHSD(m)
```